

# 18 – CRUD

Manage your data from your website

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# OUTLINE

- 1) What is the CRUD?
- 2) Routes
- 3) Read
- 4) From R allow CUD
- 5) Delete
- 6) Create
- 7) Update
- 8) INNER JOIN

# 1) WHAT IS THE CRUD?

- CRUD stands for **C**reate, **R**ead, **U**ppdate, **D**eleate

# CRUD

Create   Read   Update   Delete

Modification of the data!

# 1) WHAT IS THE CRUD?

More information: <https://www.sqlite.org/cli.html>,  
<https://www.sqlite.org/lang.html>

- **C**reate: create new data in a table:

```
INSERT INTO person (pid, fName, lName)  
VALUES (17,'Jerome','Landré');
```

- **R**ead: extract the data from one or more tables (INNER JOIN)

```
SELECT * FROM person;
```

- **U**ppdate: modify the data in a table

```
UPDATE person SET fName='Jérôme' WHERE pid=17;
```

- **D**elete: delete one or more records (rows) in a table:

```
DELETE FROM person WHERE pid=17;
```

## 2) ROUTES

- To do the CRUD on one table, we must define routes associated to each part of the CRUD, example for the table projects:

/projects : list all the projects

/projects/17 : list project number 17

/projects/new : create a new project

/projects/delete/17 : delete project 17

/projects/modify/17 : update project 17

## 2) ROUTES

- To manage errors in the CRUD process, we will go back to home.handlebars and we need to display an error message if it happens:

```
portfolio-jl-03-sessions > views > 🐛 home.handlebars > ...
1   <div>
2   |   <h1>Welcome!</h1>
3   |   <div>
4   |   |   You are visiting my personal portfolio, cool!
5   |   </div>
6   |   {{#if error}}
7   |   |   <p class="error">{{error}}</p>
8   |   {{/if}}
9   </div>
10
```

### 3) READ

The table  
projects

Table

projects

▼ Advanced

Fields Index Constraints Foreign Keys Check Constraints

Add
 Remove
 Move to top
 Move up
 Move down
 Move to bottom

| Name    | Type    | NN                                  | PK                                  | AI                                  | U                        | Default | Check | Collation |
|---------|---------|-------------------------------------|-------------------------------------|-------------------------------------|--------------------------|---------|-------|-----------|
| pid     | INTEGER | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> |         |       |           |
| pname   | TEXT    | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/> |         |       |           |
| pyear   | INTEGER | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/> |         |       |           |
| pdesc   | TEXT    | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/> |         |       |           |
| ptype   | TEXT    | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/> |         |       |           |
| pimgURL | TEXT    | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/> |         |       |           |

```

1 CREATE TABLE "projects" (
2     "pid" INTEGER,
3     "pname" TEXT NOT NULL,
4     "pyear" INTEGER NOT NULL,
5     "pdesc" TEXT NOT NULL,
6     "ptype" TEXT NOT NULL,
7     "pimgURL" TEXT NOT NULL,
8     PRIMARY KEY("pid" AUTOINCREMENT)
9 );
  
```

# 3) READ

```
app.get('/projects', (request, response) => {
  db.all('SELECT * FROM projects', (err, theProjects) => {
    if (err) {
      console.error(err.message);
      const model = { error: "Error retrieving projects from the database." }
      response.render('projects', model);
    } else {
      console.log(`---> Retrieved ${theProjects.length} projects from the database.`);
      console.log(`---> Projects: ${JSON.stringify(theProjects)}`);
      const model = { projects: theProjects };
      response.render('projects', model); // the list of projects
    }
  });
});
```

portfolio-jl-03-sessions > views > projects.handlebars > ...

```
1 <div>
2   <h1>List of Projects</h1>
3   <div>
4     <ul>
5       {{#each projects}}
6         <li>{{pname}}: {{pdesc}} Year: {{pyear}}.</li>
7       {{/each}}
8     </ul>
9     {{#if error}}
10      <p class="error">{{error}}</p>
11    {{/if}}
12  </div>
13 </div>
```



# 3) READ

```
PS C:\Users\lanjer\Desktop\lecture-sessions\portfolio-jl-03-sessions> node server
Server running on http://localhost:8080
----> Hashed password: $2b$12$mDIqfC5mOpUPLMx/ay7NOeR/FkDN4vMnWBxSP1Nox8HRqreqNN4I2
----> Retrieved 7 projects from the database.
----> Projects: [{"pid":1,"pname":"Counting people with a camera","pyear":2022,"pdesc":"The purpose of this project is to count people passing through a corridor and to know how many are in the room at a certain time.", "ptype":"Research","pimgURL":"/img/counting.png"}, {"pid":2,"pname":"Visualisation of 3D medical images","pyear":2012,"pdesc":"The project makes a 3D model of the analysis of the body of a person and displays the detected health problems. It is useful for doctors to view in 3D their patients and the evolution of a disease.", "ptype":"Research","pimgURL":"/img/medical.png"}, {"pid":3,"pname":"Multiple questions system","pyear":2021,"pdesc":"During the lockdowns in France, this project was useful to test the students online with a Quizz system.", "ptype":"Teaching","pimgURL":"/img/qcm07.png"}, {"pid":4,"pname":"Image comparison with the Local Dissimilarity Map","pyear":2020,"pdesc":"The project is about finding and quantifying the differences between two images of the same size. The applications were numerous: satellite imaging, medical imaging,...", "ptype":"Research","pimgURL":"/img/diaw02.png"}, {"pid":5,"pname":"Management system for students' internships","pyear":2012,"pdesc":"This project was about the creation of a database to manage the students' internships.", "ptype":"Teaching","pimgURL":"/img/management.png"}, {"pid":6,"pname":"Magnetic Resonance Spectroscopy","pyear":2013,"pdesc":"Analysis of signals and images from Magnetic Resonance Spectroscopy and Imaging.", "ptype":"Research","pimgURL":"/img/yu00.png"}, {"pid":7,"pname":"Signal Analysis for Detection of Epileptic Diseases","pyear":2019,"pdesc":"This project was about the detection of epileptic problems in signals.", "ptype":"Research","pimgURL":"/img/youssef00.png"}]
```

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## List of Projects

- Counting people with a camera: The purpose of this project is to count people passing through a corridor and to know how many are in the room at a certain time. Year: 2022.
- Visualisation of 3D medical images: The project makes a 3D model of the analysis of the body of a person and displays the detected health problems. It is useful for doctors to view in 3D their patients and the evolution of a disease. Year: 2012.
- Multiple questions system: During the lockdowns in France, this project was useful to test the students online with a Quizz system. Year: 2021.
- Image comparison with the Local Dissimilarity Map: The project is about finding and quantifying the differences between two images of the same size. The applications were numerous: satellite imaging, medical imaging,... Year: 2020.
- Management system for students' internships: This project was about the creation of a database to manage the students' internships. Year: 2012.
- Magnetic Resonance Spectroscopy: Analysis of signals and images from Magnetic Resonance Spectroscopy and Imaging. Year: 2013.
- Signal Analysis for Detection of Epileptic Diseases: This project was about the detection of epileptic problems in signals. Year: 2019.

## 3) READ

- A requirement for your project is that an element (for my example, a project) must be clickable to have a view on that element (that project), let's modify the code to do so.
- We want to add a link `<a href="/projects/17">` on the project name (plus `</a>`) to make it clickable and go to the route `"/projects/17"` when we click on the project name.

# 3) READ

portfolio-jl-03-sessions > views > 🍷 projects.handlebars > ...

```

1  <div>
2    <h1>List of Projects</h1>
3    <div>
4      <ul>
5        {{#each projects}}
6          <li><a href="/projects/{{pid}}">{{pname}}</a>:  {{pdesc}} Year: {{pyear}}.</li>
7        {{/each}}
8      </ul>
9      {{#if error}}
10     <p class="error">{{error}}</p>
11     {{/if}}
12   </div>
13 </div>

```

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## List of Projects

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- [Magnetic Resonance Spectroscopy](#): Analysis of signals and images from Magnetic Resonance Spectroscopy and Imaging. Year: 2013.
- [Signal Analysis for Detection of Epileptic Diseases](#): This project was about the detection of epileptic problems in signals. Year: 2019.

localhost:8080/projects/3

We need to define a route  
"/projects/:projid" in our  
"server.js" file.

### 3) READ

- We will extract the parameter from the route i.e. 17 from "/projects/17" for instance:

```
app.get('/projects/:projid', (request,response) => {
  let myPid=request.params.projid
  db.get('SELECT * FROM projects WHERE pid=?', [myPid], (err, theProject) => {
    if (err) {
      console.error(err.message);
      const model = { error: "Error retrieving the project from the database." }
      response.render('one-project', model);
    } else {
      console.log(`---> Retrieved ${theProject.length} projects from the database.`);
      console.log(`---> Project: ${JSON.stringify(theProject)}`);
      const model = { p: theProject };
      response.render('one-project', model);
    }
  });
});
```

one-project.handlebars

```
<div>
  <h1>Project details</h1>
  <div>
    <h2>{{p.pname}} - {{p.pyear}}</h2>
    <p>Type: {{p.ptype}}</p>
    <p>{{p.pdesc}}</p>
    <p></p>

    {{#if error}}
      <p class="error">{{error}}</p>
    {{/if}}
  </div>
</div>
```

# 3) READ

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## List of Projects

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- [Multiple questions system](#): During the lockdowns in France, this project was useful to test the students online with a Quizz system. Year: 2021.
- [Image comparison with the Local Dissimilarity Map](#): The project is about finding and quantifying the differences between two images of the same size. The applications were numerous: satellite imaging, medical imaging, ... Year: 2020.
- [Management system for students' internships](#): This project was about the creation of a data management system for students' internships. Year: 2012.
- [Magnetic Resonance Spectroscopy](#): Analysis of signals and images from Magnetic Resonance. Year: 2013.
- [Signal Analysis for Detection of Epileptic Diseases](#): This project was about the detection of epileptic diseases. Year: 2019.

## Project details

### Image comparison with the Local Dissimilarity Map - 2020

Type: Research

The project is about finding and quantifying the differences between two images of the same size. The applications were numerous: satellite imaging, medical imaging,...



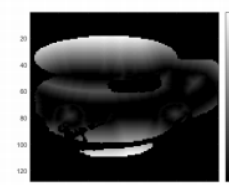
images 1 et 2



CDL images 1 et 2



images 1 et 3



CDL images 1 et 3

## 4) FROM R ALLOW CUD

- When we list all the projects, we want to allow the admin to Create, Update and Delete projects.
- We must allow the modification of the table project **ONLY** if the logged in user is an admin!
- I recall that the session is available from **ANY** file in our project because of the code:

```
//--- MAKE THE SESSION AVAILABLE IN ALL HANDLEBAR FILES AT ONCE!  
app.use((request, response, next) => {  
  response.locals.session = request.session  
  next()  
})
```

```

<div>
  <h1>List of Projects</h1>
  <div>
    {{#if session.isAdmin}}
      <p>
        <form style="display:inline" method="GET" action="/projects/new">
          <button type="submit" class="important">New project</button>
        </form>
      </p>
    {{/if}}
    <ul>
      {{#each projects}}
        <li>
          <a href="/projects/{{pid}}">{{pname}}</a>:  {{pdesc}} Year: {{pyear}}.
          {{#if ../session.isAdmin}}
            <form style="display:inline" method="GET" action="/projects/modify/{{pid}}">
              <button type="submit" class="important">Edit</button>
            </form>
            <form style="display:inline" method="POST" action="/projects/delete/{{pid}}">
              <button type="submit" class="important">X</button>
            </form>
          {{/if}}
        </li>
      {{/each}}
    </ul>
    {{#if error}}
      <p class="error">{{error}}</p>
    {{/if}}
  </div>
</div>

```



## 4) FROM R ALLOW CUD

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### List of Projects

New project

- [Counting people with a camera](#): The purpose of this project is to count people passing through a corridor and to know how many are in the room at a certain time. Year: 2022. [Edit](#) [X](#)
- [Visualisation of 3D medical images](#): The project makes a 3D model of the analysis of the body of a person and displays the detected health problems. It is useful for doctors to view in 3D their patients and the evolution of a disease. Year: 2012. [Edit](#) [X](#)
- [Multiple questions system](#): During the lockdowns in France, this project was useful to test the students online with a Quizz system. Year: 2021. [Edit](#) [X](#)
- [Image comparison with the Local Dissmilarity Map](#): The project is about finding and quantifying the differences between two images of the same size. The applications were numerous: satellite imaging, medical imaging,... Year: 2020. [Edit](#) [X](#)
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- [Magnetic Resonance Spectroscopy](#): Analysis of signals and images from Magnetic Resonance Spectroscopy and Imaging. Year: 2013. [Edit](#) [X](#)
- [Signal Analysis for Detection of Epileptic Diseases](#): This project was about the detection of epileptic problems in signals. Year: 2019. [Edit](#) [X](#)



## 5) DELETE

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### List of Projects

[New project](#)

- [Counting people with a camera](#): The purpose of this project is to count people passing through a corridor and to know how many are in the room at a certain time. Year: 2022. [Edit](#) [X](#)
- [Visualisation of 3D medical images](#): The project makes a 3D model of the analysis of the body of a person and displays the detected health problems. It is useful for doctors to view in 3D their patients and the evolution of a disease. Year: 2012. [Edit](#) [X](#)
- [Multiple questions system](#): During the lockdowns in France, this project was useful to test the students online with a Quizz system. Year: 2021. [Edit](#) [X](#)
- [Image comparison with the I](#)

```
<form style="display:inline" method="POST" action="/projects/delete/{{pid}}">
  <button type="submit" class="important">X</button>
</form>
```

This will send a POST request on the route: /projects/delete/2

## 5) DELETE

```
app.post('/projects/delete/:projid', (request,response) => {  
  let myPid=request.params.projid  
  
  if (request.session.isAdmin) {  
    db.run('DELETE FROM projects WHERE pid=?', [myPid], (err) => {  
      if (err) {  
        console.error(err.message);  
        console.log("Error deleting the project from the database.")  
        response.redirect('/');  
      } else {  
        console.log("The project was deleted.")  
        response.redirect('/')  
      }  
    })  
  } else {  
    model = { error: "You must be logged in as admin to delete a project." }  
    response.render('/login', model)  
  }  
})
```

# 5) DELETE

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## List of Projects

New project

- [Counting people with a camera](#): The purpose of this project is to count people passing through a corridor and to know how many are in the room at a certain time. Year: 2022. [Edit](#) [X](#)
- [Visualisation of 3D medical images](#): The project makes a 3D model of the analysis of the body of a person and displays the detected health problems. It is useful for doctors to view in 3D their patients and the evolution of a disease. Year: 2012. [Edit](#) [X](#)
- [Multiple questions system](#): During the lockdowns in France, this project was useful to test the students online with a Quizz system. Year: 2021. [Edit](#) [X](#)
- [Image comparison with the Local Dissmilarity Map](#): The project is about finding and quantifying the

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## Welcome!

You are visiting my personal portfolio, cool!

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## List of Projects

New project

- [Counting people with a camera](#): The purpose of this project is to count people passing through a corridor and to know how many are in the room at a certain time. Year: 2022. [Edit](#) [X](#)
- [Multiple questions system](#): During the lockdowns in France, this project was useful to test the students online with a Quizz system. Year: 2021. [Edit](#) [X](#)
- [Image comparison with the Local Dissmilarity Map](#): The project is about finding and quantifying the differences between two images of the same size. The applications were numerous: satellite imaging, medical imaging,... Year: 2020. [Edit](#) [X](#)
- [Management system for students' internships](#): This project was about the creation of a database to manage the students' internships. Year: 2012. [Edit](#) [X](#)
- [Magnetic Resonance Spectroscopy](#): Analysis of signals and images from Magnetic Resonance Spectroscopy and Imaging. Year: 2013. [Edit](#) [X](#)
- [Signal Analysis for Detection of Epileptic Deseases](#): This project was about the detection of epileptic problems in signals. Year: 2019. [Edit](#) [X](#)

## 6) CREATE

- To create a new project, we need to fill-in a form.
- Let's create the form in a form-project.handlebars file:

```
<div id="main">
  <h1>New project</h1>
  <div>
    <form action="/projects/new" method="post" id="projectform">
      <fieldset>
        <legend>Project</legend>
        <label for="name">Name:</label>
        <input type="text" name="name" id="name" />
        <br />
        <label for="desc">Description:</label>
        <textarea name="desc" id="desc" rows="4" cols="40"></textarea>
        <br />
        <label for="year">Year:</label>
        <input type="number" name="year" id="year" />
        <br />
        <label for="type">Type:</label>
        <select name="type" id="type">
          <option value="teaching">Teaching</option>
          <option value="research">Research</option>
        </select>
        <br />
        <label for="imgurl">URL:</label>
        <input type="text" name="imgurl" id="imgurl" />
        <br />
        <input type="submit" value="Create" />
      </fieldset>
    </form>
    {{#if error}} <p class="error">{{error}}</p> {{/if}}
  </div>
</div>
```

## 6) CREATE

```
app.get('/projects/new', (request,response) => {
  if (request.session.isAdmin) {
    response.render('form-project')
  } else {
    model = { error: "You must be logged in as admin to create a new project." }
    response.render('login', model)
  }
})
```

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### List of Projects

[New project](#)

- [Counting people with a camera](#): The purpose of this project is to count people passing through a corridor and to know how many are in the room at a certain time. Year: 2022. [Edit](#) [X](#)
- [Multiple questions system](#): During the lockdowns in France, this

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### New project

Project

Name:

Description:

Year:

Type:

URL:

[Create](#)

## 6) CREATE

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### New project

Project

Name:

Description:

Year:

Type:  ▼

URL:

New project created!

```
---> SESSION INFORMATION: {"cookie":{"originalMaxAge":null,"expires":null,"httpOnly":true,"path":"/"},"isLoggedIn":true,"un":"admin","isAdmin":true}
```

## 6) CREATE

- With the data coming from the form, you insert into your table:

```
app.post('/projects/new', (request, response) => {  
  // the treatment of the data received from the client form  
  console.log(`Here comes the data received from the form on the client: ${JSON.stringify(request.body)} `)  
  const { name, year, desc, type, imgurl } = request.body  
  if (request.session.isAdmin) {  
    db.run('INSERT INTO projects (pname, pyear, pdesc, ptype, pimgURL) VALUES (?, ?, ?, ?, ?)', [name, year, desc, type, imgurl], (err) => {  
      if (err) {  
        console.error(err.message);  
        const model = { error: "Error inserting the new project into the database." }  
        response.render('home', model);  
      } else {  
        console.log('New project created!')  
        response.redirect('/')  
      }  
    });  
  } else {  
    model = { error: "You must be logged in as admin to create a new project." }  
    response.render('login', model)  
  }  
})
```

## 6) CREATE

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### List of Projects

[New project](#)

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- [Magnetic Resonance Spectroscopy](#): Analysis of signals and images from Magnetic Resonance Spectroscopy and Imaging. Year: 2013. [Edit](#) [X](#)
- [Signal Analysis for Detection of Epileptic Diseases](#): This project was about the detection of epileptic problems in signals. Year: 2019. [Edit](#) [X](#)
- [My new project](#): This is my new project for 2025 in teaching. Year: 2025. [Edit](#) [X](#)



## 7) UPDATE

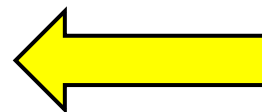
- To update a project, we need to pre-fill the project form with the values in the table for this project.
- We will re-use the form in form-project.handlebars:

```
app.get('/projects/modify/:projid', (request,response) => {  
  let myPid=request.params.projid  
  if (request.session.isAdmin) {  
    db.get('SELECT * FROM projects WHERE pid=?', [myPid], (err, theProject) => {  
      if (err) {  
        console.error(err.message);  
        const model = { error: "Error retrieving the project from the database." }  
        response.render('home', model);  
      } else {  
        const model = { p: theProject };  
        response.render('form-project', model);  
      }  
    });  
  } else {  
    model = { error: "You must be logged in as admin to modify a project." }  
    response.render('login', model)  
  }  
})
```

## 7) UPDATE

- To test if a value is equal to another in handlebar files, we must declare a helper:

```
//-----  
//--- DEFINE MIDDLEWARES  
//-----  
//--- DEFINE THE PUBLIC DIRECTORY AS STATIC  
app.use(express.static('public'))  
//--- DEFINE HANDLEBARS AS THE TEMPLATING ENGINE  
// handlebars new helper "#if (eq string1 string2)" from source: (https://stackoverflow.com/questions/34252817/handlebarsjs-check-if-a-string-is-equal-to-a-value, 2018)  
  
app.engine('handlebars',  
  engine({  
    helpers: {  
      eq (a,b) { return a==b; }  
    }  
  })  
)  
// initialize the engine to be handlebars  
app.set('view engine', 'handlebars') // set handlebars as the view engine  
app.set('views', './views') // define the views directory to be ./views
```



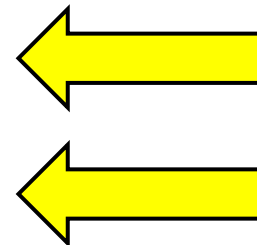
We define the eq (equality) helper function.

# 7) UPDATE

```
<div id="main">
  {{#if p.pid}}
    <h1>Modify project</h1>
  {{else}}
    <h1>New project</h1>
  {{/if}}
</div>
  {{#if p.pid}}
    <form action="/projects/modify/{{p.pid}}" method="post" id="projectform">
  {{else}}
    <form action="/projects/new" method="post" id="projectform">
  {{/if}}
    <fieldset>
      <legend>Project</legend>
      <label for="name">Name:</label>
      <input type="text" name="name" id="name"
        {{#if p.pname}} value="{{p.pname}}" {{/if}}
      />
      <br />
      <label for="desc">Description:</label>
      <textarea name="desc" id="desc" rows="4" cols="40">{{#if p.pdesc}}{{p.pdesc}}{{/if}}</textarea>
      <br />
      <label for="year">Year:</label>
      <input type="number" name="year" id="year"
        {{#if p.pyear}} value="{{p.pyear}}" {{/if}}
      />
      <br />
    </fieldset>
  </form>
</div>
```

## 7) UPDATE

```
<label for="type">Type:</label>
<select name="type" id="type">
  <option value="Teaching"
    {{#if (eq p.ptype "Teaching")}}selected{{/if}}
  >Teaching</option>
  <option value="Research"
    {{#if (eq p.ptype "Research")}}selected{{/if}}
  >Research</option>
</select>
<br />
<label for="imgurl">URL:</label>
<input type="text" name="imgurl" id="imgurl"
  {{#if p.pimgURL}} value="{{p.pimgURL}}" {{/if}}
/>
<br />
{{#if p.pid}}
  <input type="submit" value="Modify" />
{{else}}
  <input type="submit" value="Create" />
{{/if}}
</fieldset>
</form>
{{#if error}} <p class="error">{{error}}</p> {{/if}}
</div>
</div>
```



We use the eq (equality) helper function.

# 7) UPDATE

[Home](#) [Projects](#) [Skills](#) [Contact](#) Hej, admin! [Logout](#)

## List of Projects

[New project](#)

- [Counting people with a camera](#): The purpose of this project is to count people passing through a corridor and to know how many are in the room at a certain time. Year: 2022. [Edit](#) [X](#)
- [Multiple questions system](#): During the lockdowns in France, this project was useful to test the students online with a Quizz system. Year: 2021. [Edit](#) [X](#)
- [Image comparison with the Local Dissmilarity Map](#): The project is about finding and quantifying the differences between two images of the same size. The applications were numerous: satellite imaging, medical imaging,... Year: 2020. [Edit](#) [X](#)
- [Management system for students' internships](#): This project was about the creation of a database to manage the students' internships. Year: 2012. [Edit](#) [X](#)
- [Magnetic Resonance Spectroscopy](#): Analysis of signals and images from Spectroscopy and Imaging. Year: 2013. [Edit](#) [X](#)
- [Signal Analysis for Detection of Epileptic Deseases](#): This project was about epileptic problems in signals. Year: 2019. [Edit](#) [X](#)
- [My new project](#): This is my new project for 2025 in teaching. Year: 2025.

[Home](#) [Projects](#) [Skills](#) [Contact](#) Hej, admin! [Logout](#)

## Modify project

Project

Name:

Description:

Year:

Type:

URL:

[Modify](#)

# 7) UPDATE

```
app.post('/projects/modify/:projid', (request, response) => {
  let myPid=request.params.projid
  // the treatment of the data received from the client form
  console.log(`Here comes the data received from the form on the client: ${JSON.stringify(request.body)} `)
  const { name, year, desc, type, imgurl } = request.body
  if (request.session.isAdmin) {
    db.run('UPDATE projects SET pname=?, pyear=?, pdesc=?, ptype=?, pimURL=? WHERE pid=?', [name, year, desc, type, imgurl, myPid], (err) => {
      if (err) {
        console.error(err.message);
        const model = { error: "Error updating the project into the database." }
        response.render('home', model);
      } else {
        console.log('Project has been modified!')
        response.redirect('/')
      }
    });
  } else {
    model = { error: "You must be logged in as admin to create a new project." }
    response.render('login', model)
  }
})
```

# 7) UPDATE

[Home](#) [Projects](#) [Skills](#) [Contact](#) Hej, admin! [Logout](#)

## Modify project

Project

Name:

This project 3 was about the detection of epileptic problems in signals.

Description:

Year:

Type:

URL:

[Home](#) [Projects](#) [Skills](#) [Contact](#) Hej, admin! [Logout](#)

## List of Projects

- [Counting people with a camera](#): The purpose of this project is to count people passing through a corridor and to know how many are in the room at a certain time. Year: 2022.
- [Visualisation of 3D medical images](#): The project makes a 3D model of the analysis of the body of a person and displays the detected health problems. It is useful for doctors to view in 3D their patients and the evolution of a disease. Year: 2012.
- [Multiple questions system](#): During the lockdowns in France, this project was useful to test the students online with a Quizz system. Year: 2021.
- [Image comparison with the Local Dissimilarity Map](#): The project is about finding and quantifying the differences between two images of the same size. The applications were numerous: satellite imaging, medical imaging,... Year: 2020.
- [Management system for students' internships](#): This project was about the creation of a database to manage the students' internships. Year: 2012.
- [Magnetic Resonance Spectroscopy](#): Analysis of signals and images from Magnetic Resonance Spectroscopy and Imaging. Year: 2013.
- [Signal 2 Analysis for Detection of Epileptic Diseases](#): This project 3 was about the detection of epileptic problems in signals. Year: 2020.

# 7) UPDATE

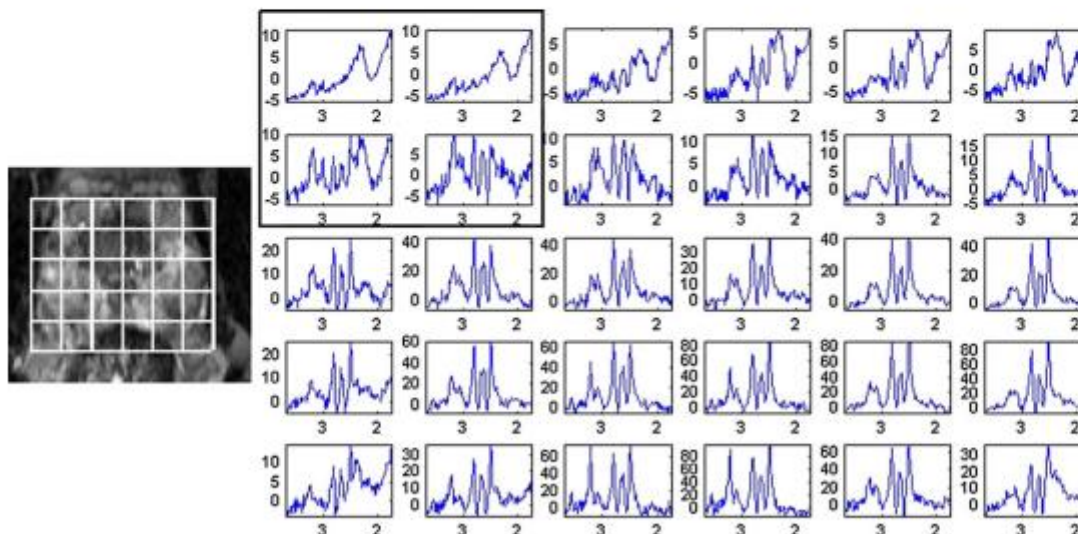
[Home](#) [Projects](#) [Skills](#) [Contact](#) Hej, admin! [Logout](#)

## Project details

### Signal 2 Analysis for Detection of Epileptic Diseases - 2020

Type: Teaching

This project 3 was about the detection of epileptic problems in signals.





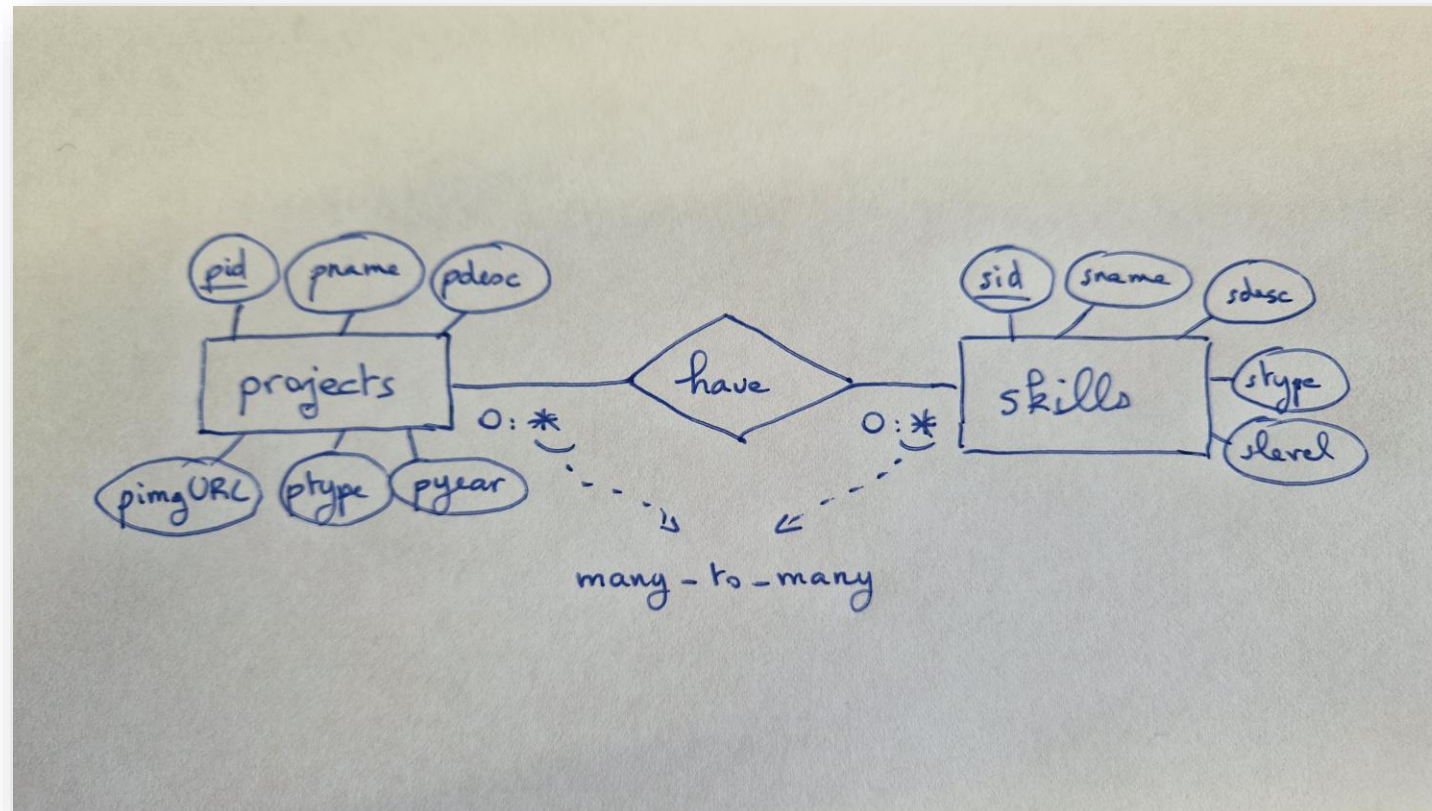
## 7) UPDATE

Be careful, the **order** of the routes is **VERY IMPORTANT!**

```
app.get('/projects', (request,response) => { ...  
  })  
  
app.get('/projects/new', (request,response) => { ...  
  })  
  
app.post('/projects/new', (request, response) => { ...  
  })  
  
app.get('/projects/:projid', (request,response) => { ...  
  })  
  
app.get('/projects/modify/:projid', (request,response) => { ...  
  })  
  
app.post('/projects/modify/:projid', (request, response) => { ...  
  })  
  
app.post('/projects/delete/:projid', (request,response) => { ...  
  })
```

## 8) INNER JOIN

- You need to JOIN tables when you need information stored in multiple tables in a SELECT:



# 8) INNER JOIN

DB Browser for SQLite - C:\Users\lanjer\Desktop\lecture-sessions\portfolio-jl-03-sessions\my-project-data.sqlite3.db

File Edit View Tools Help

New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database

Database Structure Browse Data Edit Pragma Execute SQL

Table: projects

|   | pid    | pname                                  | pyear  | pdesc                                  | ptype    | pngURL              |
|---|--------|----------------------------------------|--------|----------------------------------------|----------|---------------------|
|   | Filter | Filter                                 | Filter | Filter                                 | Filter   | Filter              |
| 1 | 1      | Counting people with a camera          | 2022   | The purpose of this project is to ...  | Research | /img/counting.png   |
| 2 | 2      | Visualisation 2 of 3D medical images   | 2020   | The project makes a 3D model of the... | Research | /img/medical.png    |
| 3 | 3      | Multiple questions system              | 2021   | During the lockdowns in France, thi... | Teaching | /img/qcm07.png      |
| 4 | 4      | Image comparison with the Local ...    | 2020   | The project is about finding and ...   | Research | /img/diaw02.png     |
| 5 | 5      | Management system for students' ...    | 2012   | This project was about the creation... | Teaching | /img/management.png |
| 6 | 6      | Magnetic Resonance Spectroscopy        | 2013   | Analysis of signals and images from... | Research | /img/yu00.png       |
| 7 | 7      | Signal 2 Analysis for Detection of ... | 2020   | This project 3 was about the ...       | Teaching | /img/yu00.png       |

DB Browser for SQLite - C:\Users\lanjer\Desktop\lecture-sessions\portfolio-jl-03-sessions\my-pro

File Edit View Tools Help

New Database Open Database Write Changes Revert Changes Undo

Database Structure Browse Data Edit Pragma Execute SQL

Table: projectshaveskills

|   | phsid  | pid    | sid    |
|---|--------|--------|--------|
|   | Filter | Filter | Filter |
| 1 | 1      | 1      | 2      |
| 2 | 2      | 1      | 8      |
| 3 | 3      | 1      | 9      |
| 4 | 4      | 2      | 2      |
| 5 | 5      | 2      | 9      |
| 6 | 6      | 3      | 1      |
| 7 | 7      | 3      | 11     |
| 8 | 8      | 3      | 12     |

DB Browser for SQLite - C:\Users\lanjer\Desktop\lecture-sessions\portfolio-jl-03-sessions\my-project-data.sqlite3.db

File Edit View Tools Help

New Database Open Database Write Changes Revert Changes Undo Open Project Save Proj

Database Structure Browse Data Edit Pragma Execute SQL

Table: skills

|    | sid    | sname        | sdesc                                  | stype                | slevel |
|----|--------|--------------|----------------------------------------|----------------------|--------|
|    | Filter | Filter       | Filter                                 | Filter               | Filter |
| 1  | 1      | PHP          | Programming with PHP on the server ... | Programming language | 4      |
| 2  | 2      | Python       | Programming with Python.               | Programming language | 4      |
| 3  | 3      | Java         | Programming with Java.                 | Programming language | 2      |
| 4  | 4      | ImageJ       | Java Framework for Image Processing.   | Framework            | 2      |
| 5  | 5      | Javascript   | Programming with Javascript on the ... | Programming language | 4      |
| 6  | 6      | Node         | Programming with Javascript on the ... | Programming language | 4      |
| 7  | 7      | Express      | A framework for programming ...        | Framework            | 4      |
| 8  | 8      | Scikit-image | A library for Image Processing with... | Library              | 3      |
| 9  | 9      | OpenCV       | A library for Image Processing with... | Library              | 4      |
| 10 | 10     | LaTeX        | A language to describe and build ...   | Description language | 5      |
| 11 | 11     | HTML         | A language to create web pages.        | Description language | 4      |
| 12 | 12     | CSS          | A language to apply styles to web ...  | Description language | 4      |
| 13 | 13     | C            | The historical language of the ...     | Programming language | 3      |
| 14 | 14     | C++          | A fast high level programming ...      | Programming language | 1      |
| 15 | 15     | SQL          | The relational database language to... | Query language       | 4      |

# 8) INNER JOIN

DB Browser for SQLite - C:\Users\lanjer\Desktop\lecture-sessions\portfolio-jl-03-sessions\my-project-data.sqlite3.db

File Edit View Tools Help

New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database

Database Structure Browse Data Edit Pragmas Execute SQL

SQL 1\*

```

1 SELECT * FROM projects
2 INNER JOIN projectshaveskills
3 ON projects.pid=projectshaveskills.pid
4 INNER JOIN skills
5 ON projectshaveskills.sid=skills.sid
6 WHERE projects.pid=2

```

|   | pid | pname                                | pyear | pdesc                                  | pptype   | pingURL          | phsid | pid | sid | sid | sname  | sdesc                                  | stype                | slevel |
|---|-----|--------------------------------------|-------|----------------------------------------|----------|------------------|-------|-----|-----|-----|--------|----------------------------------------|----------------------|--------|
| 1 | 2   | Visualisation 2 of 3D medical images | 2020  | The project makes a 3D model of the... | Research | /img/medical.png | 4     | 2   | 2   | 2   | Python | Programming with Python.               | Programming language | 4      |
| 2 | 2   | Visualisation 2 of 3D medical images | 2020  | The project makes a 3D model of the... | Research | /img/medical.png | 5     | 2   | 9   | 9   | OpenCV | A library for Image Processing with... | Library              | 4      |

Execution finished without errors.  
Result: 2 rows returned in 10ms  
At line 1:  
SELECT \* FROM projects  
INNER JOIN projectshaveskills  
ON projects.pid=projectshaveskills.pid  
INNER JOIN skills  
ON projectshaveskills.sid=skills.sid  
WHERE projects.pid=2

# 8) INNER JOIN

DB Browser for SQLite - C:\Users\lanjer\Desktop\lecture-sessions\portfolio-jl-03-sessions\my-project-data.sqlite3.db

File Edit View Tools Help

New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database

Database Structure Browse Data Edit Pragmas Execute SQL

SQL 1\*

```

1 SELECT * FROM projects
2 INNER JOIN projectshaveskills
3 ON projects.pid=projectshaveskills.pid
4 INNER JOIN skills
5 ON projectshaveskills.sid=skills.sid
6 WHERE projects.pid=2
  
```

projects projectshaveskills skills

|   | pid | pname                                | pyear | pdesc                                  | pptype   | pingURL          | phsid | pid | sid | sid | sname  | sdesc                                  | stype                | slevel |
|---|-----|--------------------------------------|-------|----------------------------------------|----------|------------------|-------|-----|-----|-----|--------|----------------------------------------|----------------------|--------|
| 1 | 2   | Visualisation 2 of 3D medical images | 2020  | The project makes a 3D model of the... | Research | /img/medical.png | 4     | 2   | 2   | 2   | Python | Programming with Python.               | Programming language | 4      |
| 2 | 2   | Visualisation 2 of 3D medical images | 2020  | The project makes a 3D model of the... | Research | /img/medical.png | 5     | 2   | 9   | 9   | OpenCV | A library for Image Processing with... | Library              | 4      |

Execution finished without errors.  
Result: 2 rows returned in 10ms  
At line 1:  
SELECT \* FROM projects  
INNER JOIN projectshaveskills  
ON projects.pid=projectshaveskills.pid  
INNER JOIN skills  
ON projectshaveskills.sid=skills.sid  
WHERE projects.pid=2

## 8) INNER JOIN

- When I will ask for the details of a project, I will get the associated skills:

```
app.get('/projects/:projid', (request, response) => {
  let myPid=request.params.projid
  db.all(`SELECT * FROM projects
    INNER JOIN projectshaveskills
    ON projects.pid=projectshaveskills.pid
    INNER JOIN skills
    ON projectshaveskills.sid=skills.sid
    WHERE projects.pid=?`, [myPid], (err, theProjectsAndSkills) => {
    if (err) {
      console.error(err.message);
      const model = { error: "Error retrieving the project from the database." }
      response.render('one-project', model);
    } else {
      console.log(`---> Retrieved ${theProjectsAndSkills.length} rows from the database.`);
      console.log(`---> Rows: ${JSON.stringify(theProjectsAndSkills)}`);
      // Extract project details (all rows have the same project info, except the skills)
      const project = theProjectsAndSkills[0];
      // Extract skills (list of all sname used in this project)
      const skills = [...new Set(theProjectsAndSkills.map(row => row.sname))];
      console.log(`---> Skills used in this project: ${skills}`);
      const model = {
        p: {
          pname: project.pname,
          pyear: project.pyear,
          ptype: project.ptype,
          pdesc: project.pdesc,
          pimURL: project.pimURL,
          skills: skills
        }
      };
      response.render('one-project', model);
    }
  });
});
```

```
<div>
  <h1>Project details</h1>
  <div>
    <h2>{{p.pname}} - {{p.pyear}}</h2>
    <p>Type: {{p.ptype}}</p>
    <p>{{p.pdesc}}</p>
    <p></p>

    <p>Skills used:</h3>
    <ul>
      {{#each p.skills}}
      <li>{{this}}</li>
      {{/each}}
    </ul>
    {{#if error}}
    <p class="error">{{error}}</p>
    {{/if}}
  </div>
</div>
```

## 8) INNER JOIN

```

---> Retrieved 3 rows from the database.
---> Rows: [{"pid":3,"pname":"Multiple questions system","pyear":2021,"pdesc":"During the lockdowns in France, this project was use
ful to test the students online with a Quizz system.","ptype":"Teaching","pimgURL":"/img/qcm07.png","phsid":6,"sid":1,"sname":"PHP"
,"sdesc":"Programming with PHP on the server side.","stype":"Programming language","slevel":4},{ "pid":3,"pname":"Multiple questions
system","pyear":2021,"pdesc":"During the lockdowns in France, this project was useful to test the students online with a Quizz sys
tem.","ptype":"Teaching","pimgURL":"/img/qcm07.png","phsid":7,"sid":11,"sname":"HTML","sdesc":"A language to create web pages.","st
ype":"Description language","slevel":4},{ "pid":3,"pname":"Multiple questions system","pyear":2021,"pdesc":"During the lockdowns in
France, this project was useful to test the students online with a Quizz system.","ptype":"Teaching","pimgURL":"/img/qcm07.png","ph
sid":8,"sid":12,"sname":"CSS","sdesc":"A language to apply styles to web pages.","stype":"Description language","slevel":4}]
---> Skills used in this project: PHP,HTML,CSS

```

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### Project details

#### Multiple questions system - 2021

Type: Teaching

During the lockdowns in France, this project was useful to test the students online with a Quizz system.

Dates/heures de début et de fin de l'épreuve :  
2021-01-15 11:15:00 --> 2021-01-15 11:45:59

1234 Jérôme Landré s1a

(1) Dans votre projet "site\_1" terminé, si on veut changer le menu du site sur toutes les pages, quel fichier doit-on modifier ? +

(2) Quel résultat est retourné par la fonction ci-dessous si on lui passe en paramètre 'Jérôme LANDRÉ' ? +

(0.5) En PHP, quelle instruction permet de faire une boucle pour parcourir un tableau associatif ? -

- ☒ foreach
- ☐ for
- ☐ if
- ☐ while

(1) En PHP, pourquoi utilise-t-on des sessions ? +

(0.5) En PHP, quel est type de \$\_POST ? +

(0.5) Quelle est la balise de fermeture du code PHP ? +

(2) On donne le formulaire ci-dessous, comment peut-on afficher le prénom de la personne dans la page de traitement traiter.php ? +

Skills used:

- PHP
- HTML
- CSS



## 8) INNER JOIN

```

---> Retrieved 2 rows from the database.
---> Rows: [{"pid":2,"pname":"Visualisation 2 of 3D medical images","pyear":2020,"pdesc":"The project makes a 3D model of the analysis of the body of a person and displays the detected health problems. It is useful for doctors to view in 3D their patients and the evolution of a disease.","ptype":"Research","pimgURL":"/img/medical.png","phsid":4,"sid":2,"sname":"Python","sdsc":"Programming with Python.","stype":"Programming language","slevel":4}, {"pid":2,"pname":"Visualisation 2 of 3D medical images","pyear":2020,"pdesc":"The project makes a 3D model of the analysis of the body of a person and displays the detected health problems. It is useful for doctors to view in 3D their patients and the evolution of a disease.","ptype":"Research","pimgURL":"/img/medical.png","phsid":5,"sid":9,"sname":"OpenCV","sdsc":"A library for Image Processing with Python.","stype":"Library","slevel":4}]
---> Skills used in this project: Python,OpenCV

```

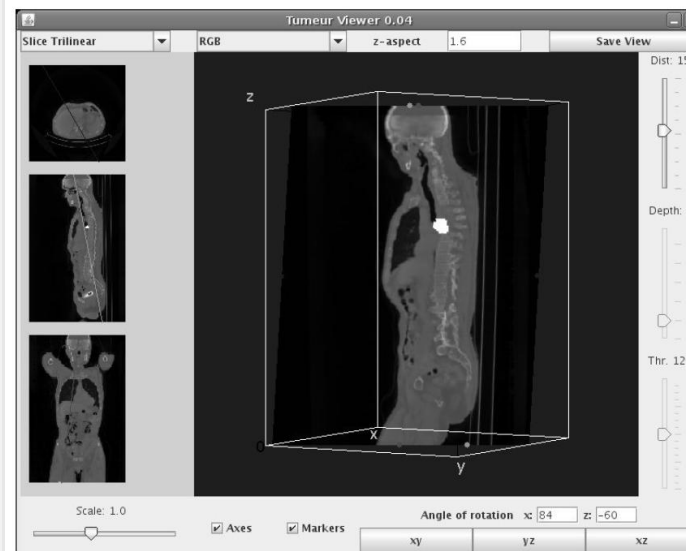
[Home](#) [Projects](#) [Skills](#) [Contact](#) [Login](#)

### Project details

#### Visualisation 2 of 3D medical images - 2020

Type: Research

The project makes a 3D model of the analysis of the body of a person and displays the detected health problems. It is useful for doctors to view in 3D their patients and the evolution of a disease.



Skills used:

- Python
- OpenCV



# ANY QUESTIONS?